

Killian Gerard

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PROFILE

Mechatronics BEng graduate with two engineering internships in live energy operations: Vitol Group on UK gas turbines and VIVO Energy (Shell licensee) on LPG and oil infrastructure in Morocco. Hands-on experience building machine-learning models in Python (battery-life prediction and high-speed-rail vibration analysis) and end-to-end CAPEX/OPEX modelling and oil-demand forecasting from fieldwork. Dual French & British national and trilingual (French, English, Spanish). Joining IE Madrid's MSc Fintech programme in September 2026.

PROFESSIONAL EXPERIENCE

- JUL–AUG 2025

Engineering Internship
VIVO Energy (Shell Licensee) · Morocco
 - Shadowed engineers across 3+ LPG and oil-infrastructure disciplines, supporting project development and management.
 - Learned end-to-end CAPEX/OPEX modelling, covering Excel structure, oil-demand forecasting and tank-reserve analytics with fail-safe systems.
 - Monitored oil-tank construction and welding under strict safety regulation; redesigned the division's engineer-onboarding deck.
- JUL–AUG 2022

Engineering Shadowing
VPI Damhead Creek, Vitol Group · UK
 - Shadowed real-time performance of Mitsubishi 701F3 gas turbines (~800 MW combined).
 - Adjusted operational output against regional daily demand; supported procurement and formal reporting.

KEY PROJECTS

- 3RD YR

ML for Remaining Useful Life of Li-ion batteries. LSTM on the Stanford/MIT dataset, MAPE < 0.07%.
- 3RD YR

Vibration analysis: high-speed-rail suspension concept with ML anomaly detection.
- 2ND YR

Line-following robot firing ping-pong balls at a target after a set distance.
- 1ST YR

IMechE robot car: drives into a wall and reverses precisely to its start position.
- 1ST YR

Scalable miniature wind-turbine aerofoil, built with JavaFoil, SolidWorks and wind-tunnel testing.

EDUCATION

- IE University, Madrid**
MSc Fintech, from Sept 2026
- Lancaster University**
Mechatronics Engineering BEng (Hons), 2023–2026
- University of Nottingham**
Physics w/ Astronomy & Foundation, 2022–2023
- Colfe's School, London**
Private secondary & sixth form, 2015–2021

LEADERSHIP & AWARDS

EPALS Vice-President · RAF (CCF) Cadet, Corporal · Duke of Edinburgh's Award, Silver & Bronze.

LANGUAGES

French & English (native), Spanish (proficient), Arabic / Portuguese / Italian (elementary).

TECHNICAL SKILLS

Python (Pandas, NumPy, TensorFlow/Keras, scikit-learn, Matplotlib, Gradio), MATLAB, C/C++ embedded. Machine learning, time-series & predictive modelling. SolidWorks, AutoCAD, JavaFoil. CAPEX/OPEX modelling.

CERTIFICATIONS

A-Levels: Maths (B), Physics (B), Spanish (A). GCSEs: 2 A*, 1 A, 1 Grade 8, 2 Grade 7s, 3 Grade 6s, 2 Grade 5s (9 subjects). Foundation Year passed with Merit. First Flight Certificate, Piper Warrior PA-28 (2019). Kaggle certificates: Python, machine-learning fundamentals, data-science fundamentals. Piano (Grade 5) and violin (Grade 4) completed.